Elizabeth Jacob

San Diego, CA (open to relocation)
elizabethhjacob@gmail.com | (619)-513 -2221

SUMMARY

Bioengineering student with hands-on experience building a wearable medical device end-to-end — integrating IMU, EMG, and PPG sensors, writing microcontroller firmware, and training validated machine-learning classifiers (90–99% accuracy) for real-time rehabilitation feedback. Comfortable across hardware, firmware, signal processing, and ML. Seeking an entry-level engineering role in medical devices or biotechnology.

PROJECTS

Smart Compression Sleeve for Physical Therapy
Senior Design Capstone | 2025 – Present

- Designed a wearable rehabilitation device integrating IMU, EMG, and PPG sensors into a compression sleeve for real-time exercise form assessment, bridging the gap between clinical systems and consumer fitness wearables.
- Trained and validated three per-exercise Random Forest classifiers (90–99% accuracy) using GroupKFold cross-validation by trial to prevent rep-level data leakage; reported results as mean ± std across 400+ labeled reps.
- Built an end-to-end real-time data pipeline: dual-IMU sensor fusion (complementary filter), Butterworth signal filtering, peak-detection rep segmentation, and feature extraction feeding the ML classifiers.
- Developed a Python WebSocket bridge for live streaming, delivering 2 Hz live form predictions plus end-of-session batch classification dispatched per exercise type.
- Wrote Arduino C++ firmware for synchronized multi-sensor acquisition (dual ICM-20948 IMUs via I^2C multiplexer, analog EMG and PPG) over a real-time USB serial link.
- Assessed regulatory and data-privacy considerations for the device, including benchmarking against FDA-cleared predicate devices and maintaining design-control documentation (Design History File).
- Coordinated ML and data workflows across hardware, firmware, and the Flutter mobile application.

Tardive Dyskinesia Risk Calculator
Independent Project | Jul 2025 – Sept 2025

- Built a machine learning–based tool to estimate patient-specific risk of tardive dyskinesia from clinical and behavioral inputs.
- Implemented data preprocessing, feature engineering, and model training using Python and scikit-learn.
- Prototyped user-facing interfaces using PySide6 and Tkinter to explore usability and workflow design.
- Used Seaborn for exploratory data analysis and visualization to guide model iteration and refinement.

EXPERIENCE

Biomaterials Innovative Lab- Student Volunteer — **December 2024- December 2025**

- Conducted cell cultivation and maintenance for biological experiments involving hepatocyte cultures.
- Prepared and tested biomaterial scaffolds for physiological and material performance analysis.
- Followed laboratory protocols to minimize contamination risk and maintain experimental consistency.

Panda Express Poway- Front of House Associate — **April 2024-December 2024**

- Worked in a fast-paced environment handling customer service, transactions, and order accuracy.
- Collaborated with team members to maintain cleanliness, efficiency, and consistent service quality.

Banner Desert Medical Center Mesa, AZ -Volunteer — **June 2020 - August 2020**

- Supported hospital operations during COVID-19 by assisting with outpatient care logistics and patient movement.
- Gained exposure to clinical workflows and patient-centered care in a high-pressure medical environment.

EDUCATION & CERTIFICATIONS

University of California, Riverside
Major: Bioengineering, Minor: Mathematics
expected graduation: June 2026

University of California, San Diego
Intro to Bioinformatics Certification
August 2023 - September 2023

TECHNICAL SKILLS

Programming & Data: Python, C++, Flask, scikit-learn, PyTorch
Engineering Tools: MATLAB, COMSOL
Data Visualization & UI: Seaborn, PySide6, Tkinter
Lab Skills: Biological and chemical laboratory techniques
Languages: English (fluent), French (intermediate)